

A smooth presentable category that is not compactly generated

Abstract

We construct a dualizable presentable stable $\mathbf{Q}$ -linear category whose coevaluation is an internal left adjoint, but which is not compactly generated. The category is the kernel of extension of scalars along a quotient of a skew Laurent algebra. Its defining ideal is flat and idempotent, and is perfect as a bimodule. A finite resolution obtained from the action on the fractional powers proves the bimodule perfectness. The classical division ring of fractions is a nonzero object of the kernel orthogonal to every compact object. This gives a counterexample to Ramzi's Conjecture 0.6.

1 Introduction

Let R be a commutative ring spectrum. An R -linear colimit-preserving functor is an *internal left adjoint* if its right adjoint is also R -linear and preserves colimits. A dualizable presentable stable R -linear category $\mathcal{C}$ is called *smooth* if its coevaluation

$$\mathrm{Mod}_R \longrightarrow \mathcal{C} \otimes_R \mathcal{C}^\vee$$

is an internal left adjoint. Equivalently, the object corresponding to $\mathrm{id}_{\mathcal{C}}$ in $\mathcal{C} \otimes_R \mathcal{C}^\vee \simeq \mathrm{Fun}_R^L(\mathcal{C}, \mathcal{C})$ is compact. Ramzi conjectures that every such category is compactly generated [2, Conjecture 0.6]. We give a counterexample for $R = H\mathbf{Q}$.

The construction uses the algebra

$$V = \bigcup_{n \geq 0} \mathbf{Q}[t^{1/2^n}]$$

and its automorphism $\sigma(t^q) = t^{2q}$, where $q \in \mathbf{Z}[1/2]_{\geq 0}$. Form the skew Laurent algebra

$$A = V[u, u^{-1}; \sigma], \quad uv = \sigma(v)u \quad (v \in V).$$

Let $I \subset A$ be the two-sided ideal spanned by the monomials $t^q u^n$ with $q > 0$. Then $A/I \simeq \mathbf{Q}[u, u^{-1}]$, with u on the right denoting the image of $u \in A$. Write $D(A)$ for the unbounded derived ∞ -category of right A -modules.

Theorem 1.1. *The category*

$$\mathcal{C} = \ker(- \otimes_A^L A/I : D(A) \longrightarrow D(A/I)) \tag{1}$$

is a dualizable presentable stable $H\mathbf{Q}$ -linear category. Its coevaluation is an internal left adjoint, and $\mathcal{C}$ is not compactly generated.

The ideal I is flat on both sides and satisfies $I \otimes_A^{\mathbf{L}} I \simeq I$. Thus tensoring with I exhibits $\mathcal{C}$ as a retract of $D(A)$ by colimit-preserving functors. We then prove that I is perfect over the enveloping algebra $A^e = A \otimes_{\mathbf{Q}} A^{\text{op}}$; this implies smoothness by lemma 2.1. The algebraic input is lemma 3.2, which expresses a quotient by infinitely many root relations as the cokernel of an endomorphism of a perfect module. Finally, the division ring of fractions of A lies in $\mathcal{C}$, whereas extension of scalars to it annihilates every compact object. This proves the failure of compact generation.

All categorical tensor products below are taken in presentable stable linear categories. We identify $\mathbf{Q}$ -linear categories with $H\mathbf{Q}$ -linear categories. For a ring Λ , $\text{Perf}(\Lambda)$ denotes the compact objects of $D(\Lambda)$, represented by bounded complexes of finitely generated projective right modules. Bimodules over a k -algebra Λ are regarded as left modules over $\Lambda \otimes_k \Lambda^{\text{op}}$, with action $(a \otimes b^{\text{op}})m = amb$; perfectness for left modules has the corresponding meaning. Background on dualizable presentable categories is given in [1, Section 1.4] and [2, Section 1.3].

2 The kernel category

We first relate smoothness of a retract of a module category to perfectness of its projection bimodule.

Lemma 2.1. *Let k be a field, let Λ be a k -algebra, and let $\mathcal{E}$ be a presentable stable k -linear category. Suppose that*

$$i : \mathcal{E} \rightleftarrows D(\Lambda) : p, \quad i \dashv p, \quad pi \simeq \text{id}_{\mathcal{E}},$$

is a k -linear adjunction in which both functors preserve colimits. If the bimodule representing ip is perfect over $\Lambda \otimes_k \Lambda^{\text{op}}$, then $\mathcal{E}$ is dualizable and smooth over k .

Proof. The functors i and p exhibit $\mathcal{E}$ as a retract of $D(\Lambda)$ in presentable k -linear categories. The latter is generated by the compact object Λ , so the retract criterion [2, Theorem 1.49] gives dualizability of $\mathcal{E}$.

Dualizing the adjunction gives $p^{\vee} \dashv i^{\vee}$ with $i^{\vee} p^{\vee} \simeq \text{id}_{\mathcal{E}^{\vee}}$. Hence

$$F = i \otimes p^{\vee} : \mathcal{E} \otimes_k \mathcal{E}^{\vee} \longrightarrow D(\Lambda) \otimes_k D(\Lambda)^{\vee}$$

has right adjoint $p \otimes i^{\vee}$. The unit of this adjunction is the tensor product of the units of $i \dashv p$ and $p^{\vee} \dashv i^{\vee}$, both of which are equivalences. Thus F is fully faithful and preserves colimits.

Let P be the bimodule representing ip . Under the equivalence between kernels and colimit-preserving functors, F sends a functor H to iHp and therefore sends $\text{id}_{\mathcal{E}}$ to P . The right adjoint to the coevaluation of $\mathcal{E}$ is consequently

$$K \longmapsto \text{RHom}_{\Lambda \otimes_k \Lambda^{\text{op}}}(P, F(K)).$$

Since P is perfect, this functor can be written as

$$\text{RHom}_{\Lambda \otimes_k \Lambda^{\text{op}}}(P, \Lambda \otimes_k \Lambda^{\text{op}}) \otimes_{\Lambda \otimes_k \Lambda^{\text{op}}}^{\mathbf{L}} F(K).$$

It preserves colimits and is k -linear, as required. $\square$

We apply the lemma to the algebra A and the ideal I of the introduction.

Proposition 2.2. *The category $\mathcal{C}$ in (1) is presentable, stable, and dualizable over $H\mathbf{Q}$. Its inclusion $i : \mathcal{C} \hookrightarrow \mathbf{D}(A)$ has a colimit-preserving right adjoint*

$$p(M) = M \otimes_A^{\mathbf{L}} I.$$

Moreover, under this inclusion,

$$\mathcal{C}^\omega = \mathcal{C} \cap \text{Perf}(A). \quad (2)$$

Proof. Let $\mathfrak{m} \subset V$ be the ideal generated by the positive powers of t . Each such power is the square of another positive power, so $\mathfrak{m}^2 = \mathfrak{m}$. Also,

$$\mathfrak{m} = \bigcup_{n \geq 0} t^{1/2^n} V$$

is a filtered union of free V -modules of rank one, hence is flat. Since $\sigma(\mathfrak{m}) = \mathfrak{m}$, we have $I = \mathfrak{m}A = A\mathfrak{m}$. The identifications

$$I \simeq \mathfrak{m} \otimes_V A \quad \text{as a right } A\text{-module}, \quad I \simeq A \otimes_V \mathfrak{m} \quad \text{as a left } A\text{-module}$$

show that I is flat on both sides.

Tensoring $I \hookrightarrow A$ with the flat module I shows that the multiplication map $I \otimes_A I \rightarrow I$ is injective. Its image is $I^2 = I$. It follows that

$$I \otimes_A^{\mathbf{L}} I \simeq I, \quad I \otimes_A^{\mathbf{L}} A/I = 0, \quad A/I \otimes_A^{\mathbf{L}} A/I \simeq A/I. \quad (3)$$

For the second identity, ordinary tensor product is $I/I^2 = 0$; the third follows by tensoring the triangle $I \rightarrow A \rightarrow A/I$.

The functor $-\otimes_A^{\mathbf{L}} A/I$ is accessible, exact, and colimit-preserving. Its kernel is therefore a presentable stable subcategory whose colimits are computed in $\mathbf{D}(A)$. For every $M \in \mathbf{D}(A)$ there is a triangle

$$M \otimes_A^{\mathbf{L}} I \longrightarrow M \longrightarrow M \otimes_A^{\mathbf{L}} A/I. \quad (4)$$

By (3), its first term lies in $\mathcal{C}$, and the first arrow is an equivalence when $M \in \mathcal{C}$.

For $X \in \mathcal{C}$ and $M \in \mathbf{D}(A)$, extension and restriction of scalars give

$$\text{RHom}_A(X, M \otimes_A^{\mathbf{L}} A/I) \simeq \text{RHom}_{A/I}(X \otimes_A^{\mathbf{L}} A/I, M \otimes_A^{\mathbf{L}} A/I) = 0.$$

Applying $\text{RHom}_A(X, -)$ to (4) proves $i \dashv p$. Both functors preserve colimits and $pi \simeq \text{id}_{\mathcal{C}}$, so the same retract criterion used in lemma 2.1 proves dualizability.

Since p preserves filtered colimits, its left adjoint i sends compact objects to compact objects. Conversely, an object of $\mathcal{C}$ that is compact in $\mathbf{D}(A)$ is compact in $\mathcal{C}$, because i is fully faithful and preserves colimits. This proves (2). $\square$

3 Perfect bimodules

Proposition 3.1. *The bimodules A , A/I , and I are perfect over A^e .*

The proof uses an automorphism to encode an increasing family of relations in a single endomorphism. We first record the resulting finite presentation in the derived category.

Lemma 3.2 (Skew telescope). *Let S be a ring, let α be an automorphism of S , and set $T = S[d, d^{-1}; \alpha]$. Let $J \subset S$ be a left ideal such that $\alpha(J) \subset J$ and S/J is a perfect left S -module. Put*

$$J_\infty = \bigcup_{r \geq 0} \alpha^{-r}(J).$$

Then S/J_∞ , with d acting through α , is a perfect left T -module.

Proof. The union defining J_∞ is increasing, and $\alpha(J_\infty) = J_\infty$, so the stated T -action is defined. The induced module

$$N = T \otimes_S S/J = T/TJ$$

is perfect over T : the right S -module T is free, and induction takes a finite projective resolution to one over T .

Right multiplication by d^{-1} induces a left T -linear map $F : N \rightarrow N$, since

$$jd^{-1} = d^{-1}\alpha(j) \in TJ \quad (j \in J).$$

As a left S -module,

$$N \simeq \bigoplus_{n \in \mathbf{Z}} S/\alpha^n(J).$$

In this description F takes the summand indexed by n to the one indexed by $n - 1$ by the quotient map. The map $1 - F$ is injective: for a nonzero vector of finite support, its component at the largest nonzero index is unchanged by $1 - F$.

The cokernel is the colimit of these quotient maps, namely S/J_∞ . More explicitly, the augmentation takes a finite sum $\sum_n s_n d^n$ to $\sum_n s_n$ modulo J_∞ . Since $d(s_n d^n) = \alpha(s_n) d^{n+1}$, it gives precisely the asserted d -action. Thus there is an exact sequence of left T -modules

$$0 \longrightarrow N \xrightarrow{1-F} N \longrightarrow S/J_\infty \longrightarrow 0. \quad (5)$$

Its last term is the cofiber of a map between perfect modules. $\square$

Proof of proposition 3.1. Identify

$$S = V \otimes_{\mathbf{Q}} V = \mathbf{Q}[x^{1/2^\infty}, y^{1/2^\infty}],$$

where the last expression denotes the union of the polynomial rings $\mathbf{Q}[x^{1/2^n}, y^{1/2^n}]$. Let α double the exponents of both variables, and let $T = S[d, d^{-1}; \alpha]$. Inside A^e , take

$$x^q = t^q \otimes 1, \quad y^q = 1 \otimes (t^q)^{\text{op}}, \quad d = u \otimes (u^{-1})^{\text{op}}.$$

Conjugation by d doubles the exponents of both x and y . These formulas identify T with a subalgebra of A^e .

We first show that V and $\mathbf{Q}$ are perfect left T -modules, where d acts on V by σ and on $\mathbf{Q}$ by the identity. Each ring $\mathbf{Q}[x^{1/2^n}, y^{1/2^n}]$ is free over $\mathbf{Q}[x, y]$, with basis

$$x^{a/2^n} y^{b/2^n}, \quad 0 \leq a, b < 2^n.$$

Consequently S is flat over $\mathbf{Q}[x, y]$, and the Koszul resolutions for $x - y$ and for (x, y) remain exact after extension to S . In particular, $S/(x - y)$ and $S/(x, y)$ are perfect over S .

For $J = (x - y)$, the inclusion $\alpha(J) \subset J$ follows from $x^2 - y^2 = (x - y)(x + y)$. Moreover,

$$\alpha^{-r}(J) = (x^{1/2^r} - y^{1/2^r}), \quad S/J_\infty \simeq V.$$

Indeed, the map to V identifies $x^{1/2^r}$ and $y^{1/2^r}$ with $t^{1/2^r}$. Every element lies in a finite polynomial stage, where its kernel is generated by the difference of those two roots. Lemma 3.2 therefore proves that V is perfect over T .

For $J = (x, y)$, the same lemma applies and

$$\alpha^{-r}(J) = (x^{1/2^r}, y^{1/2^r}), \quad S/J_\infty \simeq \mathbf{Q}.$$

Thus $\mathbf{Q}$ is also perfect over T .

It remains to identify the induced bimodules. Comparing the two Laurent degrees in A^e gives the right T -module decomposition

$$A^e = \bigoplus_{n \in \mathbb{Z}} (u \otimes 1)^n T.$$

For example, a monomial in the two Laurent generators with degrees (a, b) lies in $(u \otimes 1)^{a+b} T$, since d has degrees $(1, -1)$. Moving a coefficient past $(u \otimes 1)^n$ applies an automorphism of S , so the displayed summands are free of rank one.

The bimodule action defines maps

$$A^e \otimes_T V \longrightarrow A, \quad A^e \otimes_T \mathbf{Q} \longrightarrow A/I. \quad (6)$$

The first sends $(a \otimes b^{\text{op}}) \otimes v$ to avb . On the free summand indexed by n , it is

$$(u \otimes 1)^n \otimes v \longmapsto u^n v = \sigma^n(v) u^n,$$

which is a bijection onto the n th Laurent summand of A . The second map is given by the same formula in the quotient; it sends $(u \otimes 1)^n \otimes c$ to cu^n . Hence both maps in (6) are isomorphisms.

Since A^e is free as a right T -module, these maps also compute derived induction. Induction preserves perfect objects, so A and A/I are perfect over A^e . The triangle $I \rightarrow A \rightarrow A/I$ then proves the assertion for I . $\square$

Corollary 3.3. *The category $\mathcal{C}$ is smooth over $H\mathbf{Q}$.*

Proof. By proposition 2.2, the projection ip is represented by the bimodule I . Apply lemma 2.1 and proposition 3.1. $\square$

4 Failure of compact generation

Proposition 4.1. *The category $\mathcal{C}$ is not compactly generated.*

We will produce a nonzero object of $\mathcal{C}$ orthogonal to all its compact objects. This object is the division ring of fractions of A . The point requiring proof is that every perfect A -complex annihilated by extension to A/I is also annihilated by extension to this division ring.

4.1 The division ring of fractions

Lemma 4.2. *The algebra A is a left and right Ore domain. Its classical division ring of fractions E is flat on both sides over A and satisfies*

$$E \otimes_A^{\mathbf{L}} A/I = 0.$$

In particular, E is a nonzero object of $\mathcal{C}$.

Proof. The automorphism σ extends to $\text{Frac}(V)$. The skew polynomial ring over this field admits left and right division algorithms, obtained by canceling leading terms using σ and σ^{-1} . Hence its skew Laurent ring $\text{Frac}(V)[u, u^{-1}; \sigma]$ is a left and right Ore domain; let E be its division ring of fractions.

We check that denominators can be chosen in A . Write a skew Laurent polynomial as

$$f = \sum_n (a_n/b_n)u^n, \quad a_n, b_n \in V, \quad b_n \neq 0,$$

with finitely many nonzero terms. Multiplication on the left by $\prod_n b_n$ clears its coefficients. Multiplication on the right by

$$s = \prod_n \sigma^{-n}(b_n)$$

also clears them, because the coefficient of u^n in fs is $(a_n/b_n)\sigma^n(s) \in V$. The same choices, with the products taken over all coefficients, clear any finite family simultaneously.

For a right fraction $fg^{-1} \in E$, simultaneous right clearing gives

$$fg^{-1} = (fs)(gs)^{-1}, \quad fs, gs \in A.$$

The analogous left clearing gives a left fraction over A . Clearing common multiples in the skew Laurent ring gives common multiples in A as well. Thus A is Ore on both sides, and E is its classical division ring of fractions.

As a right A -module, E is the filtered union of the submodules $a^{-1}A$ for $0 \neq a \in A$; common left multiples make this union filtered. Each is free of rank one. Likewise, as a left A -module, E is the filtered union of the free modules Aa^{-1} . Thus E is flat on both sides.

Finally t is invertible in E and zero in A/I . For $e \in E$ and $b \in A/I$,

$$e \otimes b = (et^{-1}) \otimes tb = 0.$$

Flatness gives the derived vanishing, and $E \neq 0$ because it contains the domain A . $\square$

4.2 Finite matrices and perfect complexes

Lemma 4.3. *For every $X \in M_r(I)$, the matrix $1 - X$ is invertible in $M_r(E)$. More generally, every matrix in $M_r(A)$ whose image in $M_r(A/I)$ is invertible becomes invertible over E .*

Proof. The monomials $t^q u^n$, with $q \in \mathbf{Z}[1/2]_{\geq 0}$ and $n \in \mathbf{Z}$, form a $\mathbf{Q}$ -basis of A , with multiplication

$$(t^q u^n)(t^r u^m) = t^{q+2^r r} u^{n+m}. \quad (7)$$

Give a monomial the weight

$$w(t^q u^n) = 2^{-n} q.$$

If $a = t^q u^n$ has $q > 0$ and $b = t^r u^m$, then

$$w(ab) = w(b) + 2^{-(n+m)} q > w(b). \quad (8)$$

Regard A^r as a right module of column vectors, with matrices acting on the left. A nonzero vector $v \in A^r$ contains finitely many monomials, so their weights have a minimum w_0 . Every monomial in each entry of X has positive t -exponent. Equation (8) shows that every term in Xv has weight strictly larger than w_0 . Combining terms can only remove monomials. Therefore $v \neq Xv$, and $1 - X$ is injective on A^r .

If $1 - X$ had a nonzero kernel vector in E^r , a common right denominator would carry it to a nonzero kernel vector in A^r . Thus $1 - X$ is injective on the finite-dimensional right E -vector space E^r , and hence invertible.

For the last assertion, let $M \in M_r(A)$ and lift the inverse of its image in $M_r(A/I)$ to $N \in M_r(A)$. Then $NM = 1 - X$ for some $X \in M_r(I)$. The first assertion makes NM invertible over E , so M is injective on E^r and is therefore invertible. $\square$

Lemma 4.4. *If $P \in \text{Perf}(A)$ and $P \otimes_A^{\mathbf{L}} A/I = 0$, then $P \otimes_A^{\mathbf{L}} E = 0$.*

Proof. Represent P by a bounded cochain complex of finitely generated projective right A -modules. Its extension to A/I is a bounded acyclic complex of projective modules, hence is contractible: starting at the last nonzero term, successive surjections split and the argument proceeds by induction. Choose a contracting homotopy $h_{A/I}$.

This homotopy lifts degree by degree to an A -linear map h of degree -1 . Indeed, compose each component of $h_{A/I}$ with $P^j \rightarrow P^j \otimes_A A/I$ and lift it along $P^{j-1} \twoheadrightarrow P^{j-1} \otimes_A A/I$, using the projectivity of P^j . Put

$$U = dh + hd.$$

Then U is a chain endomorphism, and $U \otimes_A A/I = 1$.

For each degree, choose a finitely generated projective complement to P^j and extend U^j by the identity on that complement. On the resulting finite free module, this is a matrix congruent to the identity modulo I . By lemma 4.3, it is invertible over E . Its restriction to $P^j \otimes_A E$ is therefore invertible as well. It follows that $U \otimes_A E$ is a chain isomorphism.

After extension to E , its inverse commutes with the differential. Suppressing the extension of scalars in the notation, we obtain

$$d(hU^{-1}) + (hU^{-1})d = (dh + hd)U^{-1} = 1.$$

Thus $P \otimes_A^{\mathbf{L}} E$ is contractible. □

Proof of proposition 4.1. By proposition 2.2, every compact object $c \in \mathcal{C}$ is perfect as an A -complex. Since $c \otimes_A^{\mathbf{L}} A/I = 0$, lemma 4.4 gives $c \otimes_A^{\mathbf{L}} E = 0$. Consequently

$$\text{RHom}_{\mathcal{C}}(c, E) \simeq \text{RHom}_A(c, E) \simeq \text{RHom}_E(c \otimes_A^{\mathbf{L}} E, E) = 0.$$

By lemma 4.2, E is a nonzero object of $\mathcal{C}$. It is therefore a nonzero object orthogonal to every compact object and all of its shifts, so compact objects do not generate $\mathcal{C}$. □

Proof of theorem 1.1. Dualizability and presentability are proved in proposition 2.2, smoothness in corollary 3.3, and failure of compact generation in proposition 4.1. □

Remark 4.5. The category $\mathcal{C}$ contains nonzero compact objects. For example, let

$$P = \text{cofib}(A \xrightarrow{L_{1-t}} A),$$

where L_{1-t} denotes left multiplication by $1-t$, a map of right A -modules. This is a perfect complex, and after extension to A/I the arrow is the identity. Thus $P \in \mathcal{C}^\omega$ by (2). The augmentation $A \rightarrow \mathbf{Q}$ defined by $t^q \mapsto 1$ and $u \mapsto 1$ kills $(1-t)A$ but does not kill 1, so $A/(1-t)A \neq 0$ and $P \neq 0$.

Remark 4.6. The bimodule finiteness used in proposition 3.1 does not imply one-sided finiteness of I . In fact, I is not finitely generated as a right A -module. The monomials in any finite family of its elements have t -exponents bounded below by some $\varepsilon > 0$. By (7), right multiplication by A cannot decrease those exponents. Such a family cannot generate t^q for $0 < q < \varepsilon$.

The example disproves [2, Conjecture 0.6]. The variants requiring both evaluation and coevaluation to be internal left adjoints, or requiring invertibility, remain outside this conclusion.

A Explicit finite resolutions

We give the free resolutions underlying proposition 3.1. Use the ring T and variables x, y, d from that proof. Matrices act on row vectors on the right, so they represent maps of left free T -modules. The relations are

$$xd^{-1} = d^{-1}x^2, \quad yd^{-1} = d^{-1}y^2.$$

The module V has the resolution

$$0 \longrightarrow T \xrightarrow{D_2} T^2 \xrightarrow{D_1} T \longrightarrow V \longrightarrow 0,$$

where

$$D_1 = \begin{pmatrix} 1 - d^{-1} \\ x - y \end{pmatrix}, \quad D_2 = \begin{pmatrix} -(x - y) & 1 - d^{-1}(x + y) \end{pmatrix}.$$

To obtain it, induce the Koszul resolution for $x - y$ from S to T . Right multiplication by d^{-1} on $T/T(x - y)$ lifts to the chain map with components

$$F_0 = d^{-1}, \quad F_1 = d^{-1}(x + y),$$

because $F_1(x - y) = (x - y)F_0$. The displayed resolution is the mapping cone of $1 - F_\bullet$.

For $\mathbf{Q}$, the corresponding resolution is

$$0 \longrightarrow T \xrightarrow{D_3} T^3 \xrightarrow{D_2} T^3 \xrightarrow{D_1} T \longrightarrow \mathbf{Q} \longrightarrow 0,$$

where

$$D_1 = \begin{pmatrix} 1 - d^{-1} \\ x \\ y \end{pmatrix}, \quad D_2 = \begin{pmatrix} -x & 1 - d^{-1}x & 0 \\ -y & 0 & 1 - d^{-1}y \\ 0 & -y & x \end{pmatrix},$$

$$D_3 = \begin{pmatrix} y & -x & 1 - d^{-1}xy \end{pmatrix}.$$

Here the Koszul differentials are $\begin{pmatrix} x \\ y \end{pmatrix}$ and $\begin{pmatrix} -y & x \end{pmatrix}$. Right multiplication by d^{-1} on $T/T(x, y)$ lifts to

$$F_0 = d^{-1}, \quad F_1 = \text{diag}(d^{-1}x, d^{-1}y), \quad F_2 = d^{-1}xy.$$

The identities $xd^{-1} = d^{-1}x^2$ and $yd^{-1} = d^{-1}y^2$ give the chain-map conditions, and the displayed differentials again form the cone of $1 - F_\bullet$.

In both cases the induced Koszul complex has homology only in degree zero. On that homology, $1 - F$ is injective by lemma 3.2, with cokernel V or $\mathbf{Q}$, respectively. The long exact sequence of a mapping cone proves exactness of the two resolutions. Extension of scalars along $T \rightarrow A^e$ then gives finite free bimodule resolutions of A and A/I .

References

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